



DEEP CONVOLUTION TABLES : DEEP LEARNING WITHOUT CONVOLUTIONS

MATHEMATICS FOR COMPUTING - 2

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ABSTRACT

- This project explores and compares three deep learning architectures - Convolutional Neural Networks (CNNs), Fully Connected Neural Networks (FCNNs), and Convolution Tables (CTs), with our main focus on CTs. The primary dataset used is **CIFAR-10**, and additional experiments on datasets like **MNIST** and **SVHN** in **MATLAB** and **Jupyter Notebook** mainly.
- We have also implemented CT (CIFAR-10) in marimo notebook and this presentation also includes **UID-based result** developed using **Flask (Python)**, integrated with a **SQLite** database and implemented via **VS Code**.
- **Weka**-based classification models are also included in the main presentation.

INTRODUCTION

- This project explores the innovative use of Convolutional Tables (CTs) as a substitute for traditional convolutional layers in deep learning architectures.
- CT layers employ a decision-tree-like approach, using simple binary (yes/no) choices and a voting mechanism to identify features within the data, thereby eliminating the need for complex convolutional filters.
- The primary objective is to evaluate whether CT layers can achieve comparable accuracy to Convolutional Neural Networks (CNNs) while significantly reducing computational demands.
- Our project focuses into the advantages and limitations of this approach, assessing its potential for efficient deep learning.

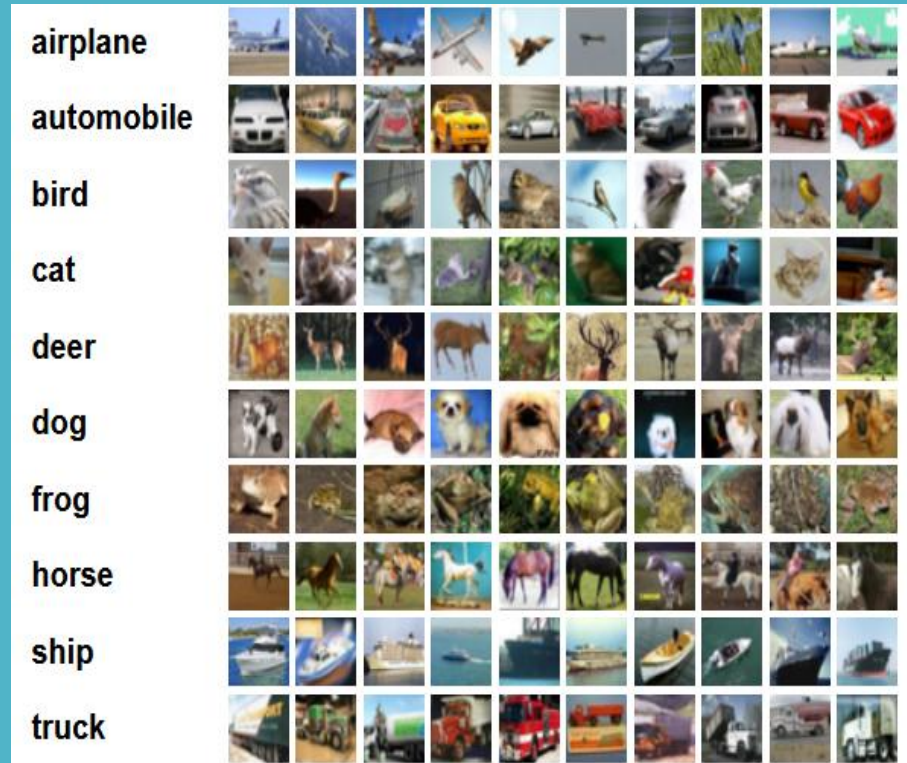
DATASET OVERVIEW

➤ CIFAR-10 Dataset

- ❑ Instances: 60,000 images
- ❑ Image Size: 32×32 pixels
- ❑ Channels: 3 (RGB)
- ❑ Classes: 10
- ❑ Training Images: 50,000
- ❑ Test Images: 10,000
- ❑ Training Batches: 5 (10,000 images each)
- ❑ Test Batch: 1 (1,000 images per class)

➤ Class Labels:

- ❑ Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck

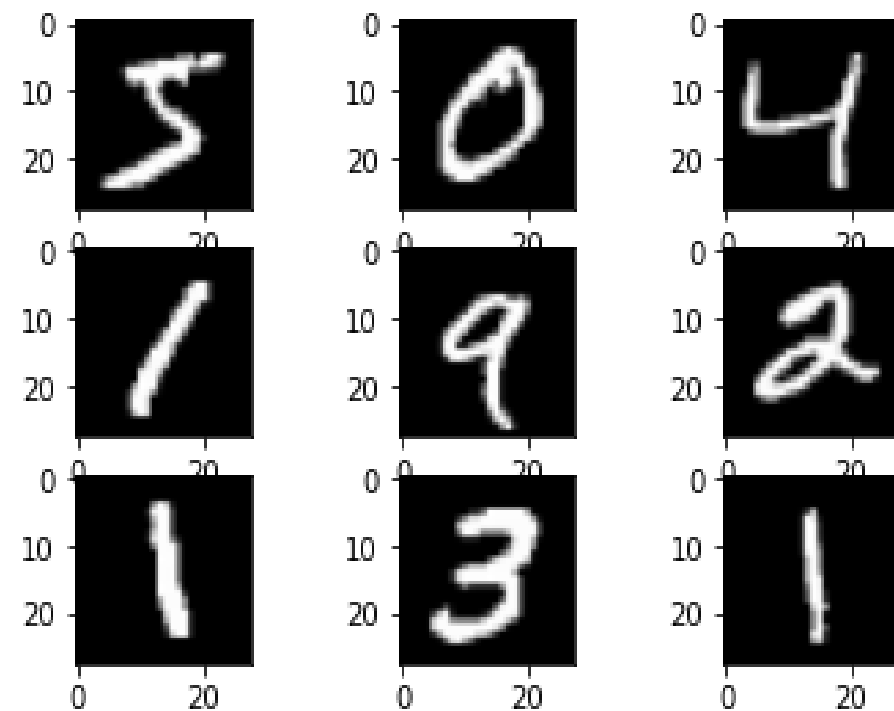


➤ MNIST dataset

- ❑ Instances: 70,000 images
- ❑ Image Size: 28×28 pixels
- ❑ Channels: 1 (Grayscale)
- ❑ Classes: 10 (Digits 0–9)
- ❑ Training Images: 60,000
- ❑ Test Images: 10,000
- ❑ Pixel Range: 0 to 255 (intensity)

➤ Class Labels

- ❑ Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

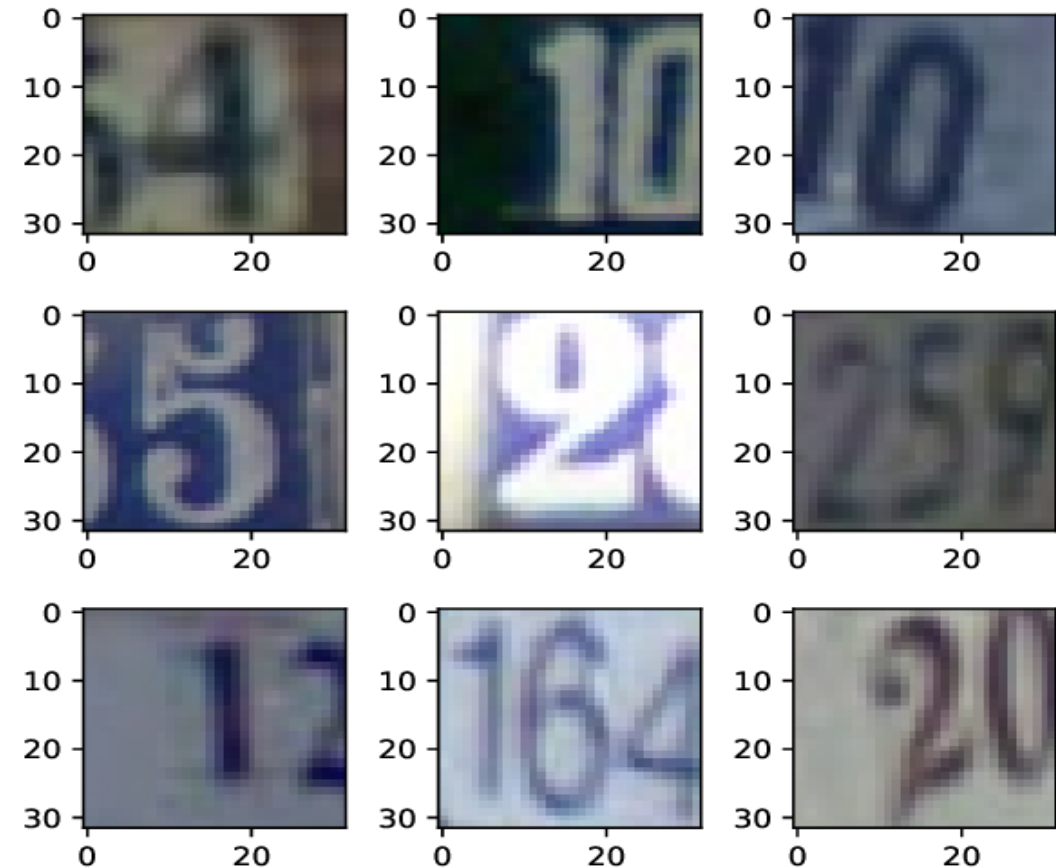


SVHN dataset

- ❑ Instances: 600,000+ images
- ❑ Image Type: Real-world house numbers
- ❑ Source: Google Street View
- ❑ Image Size: 32×32 pixels
- ❑ Channels: 3 (RGB)
- ❑ Classes: 10 (Digits 0–9)

➤ Labels

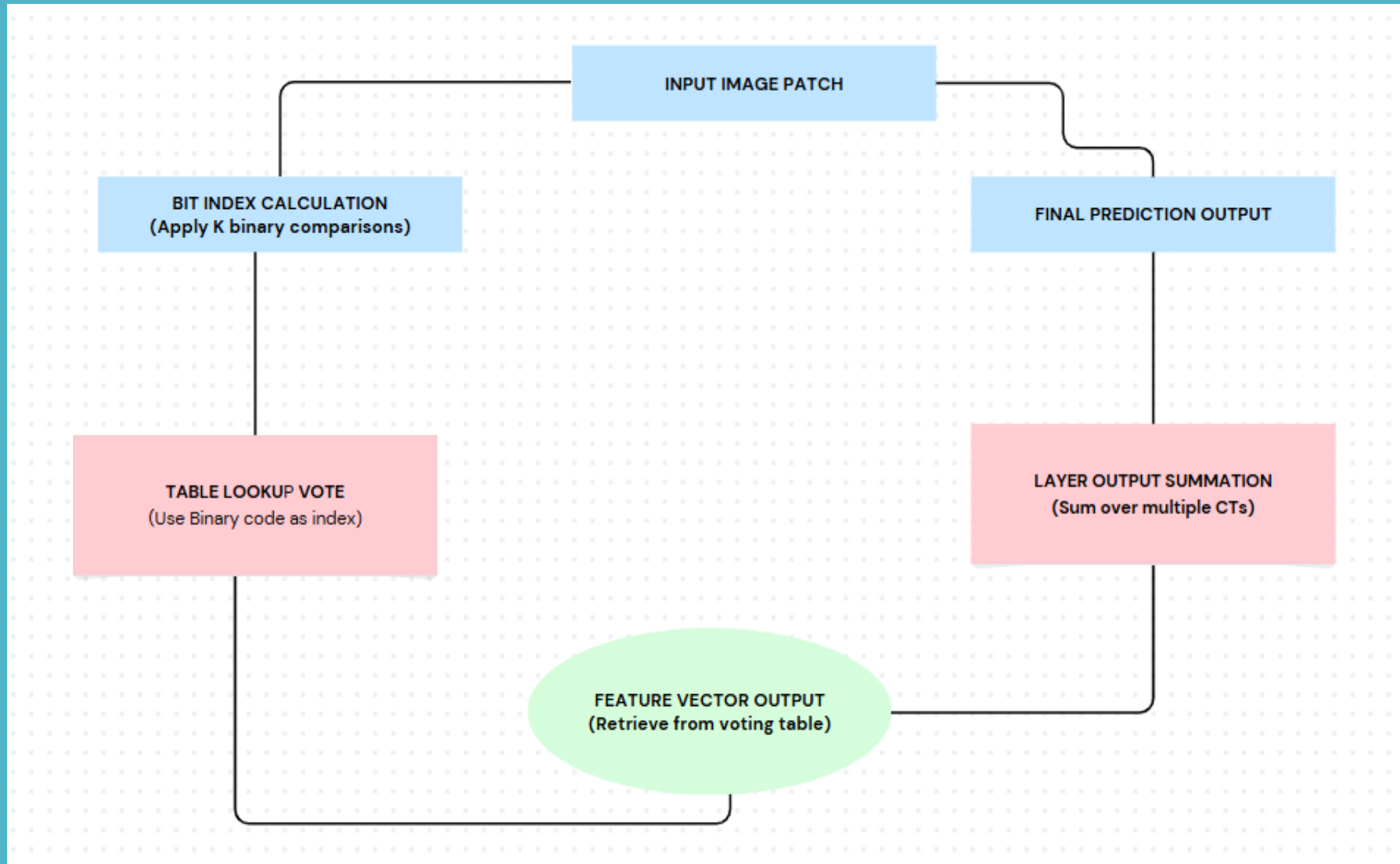
- ❑ Digit '1' → Label 1
- ❑ Digit '9' → Label 9
- ❑ Digit '0' → Label 10



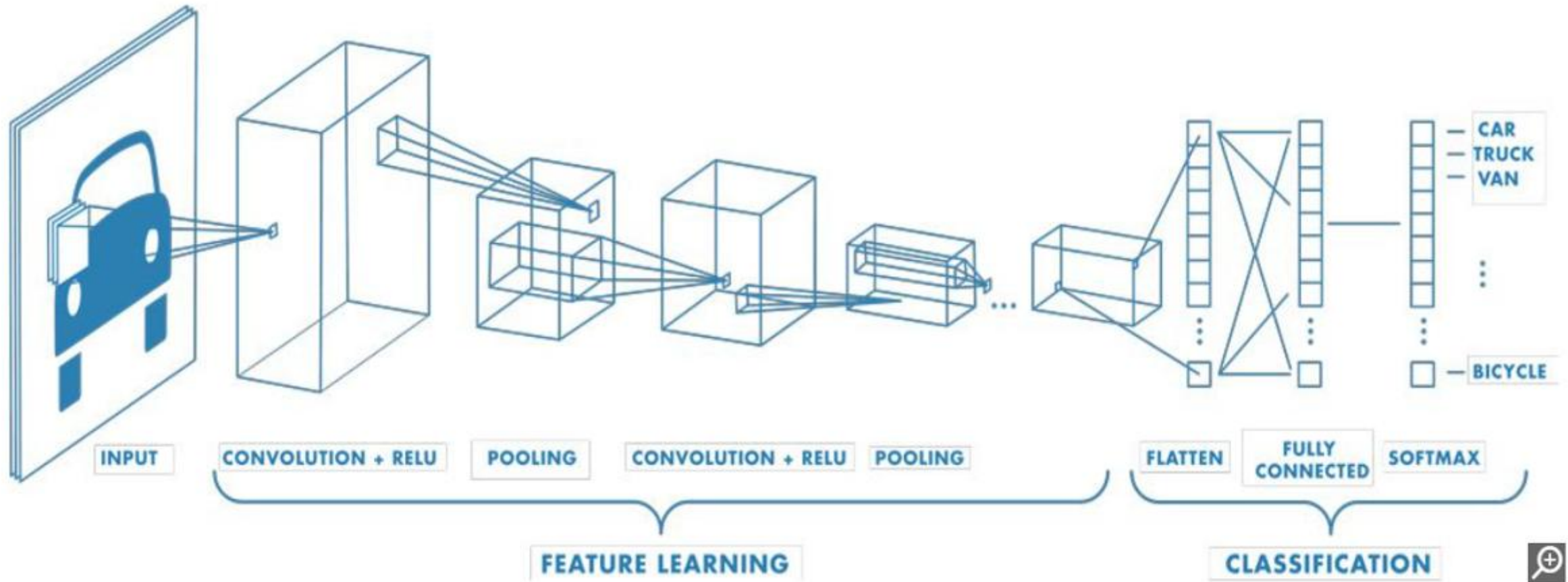
WHAT ARE CONVOLUTION TABLES ?

- The **Convolutional Table (CT)** approach replaces dot-product operations in CNNs with memory-efficient voting mechanisms.
- Each image patch is encoded into a binary index using simple comparisons, which retrieves outputs from pre-trained tables.
- These outputs are summed across multiple tables to form feature maps, like CNN layers.
- CTs offer significantly faster inference on CPUs by trading compute-heavy operations for memory lookups.

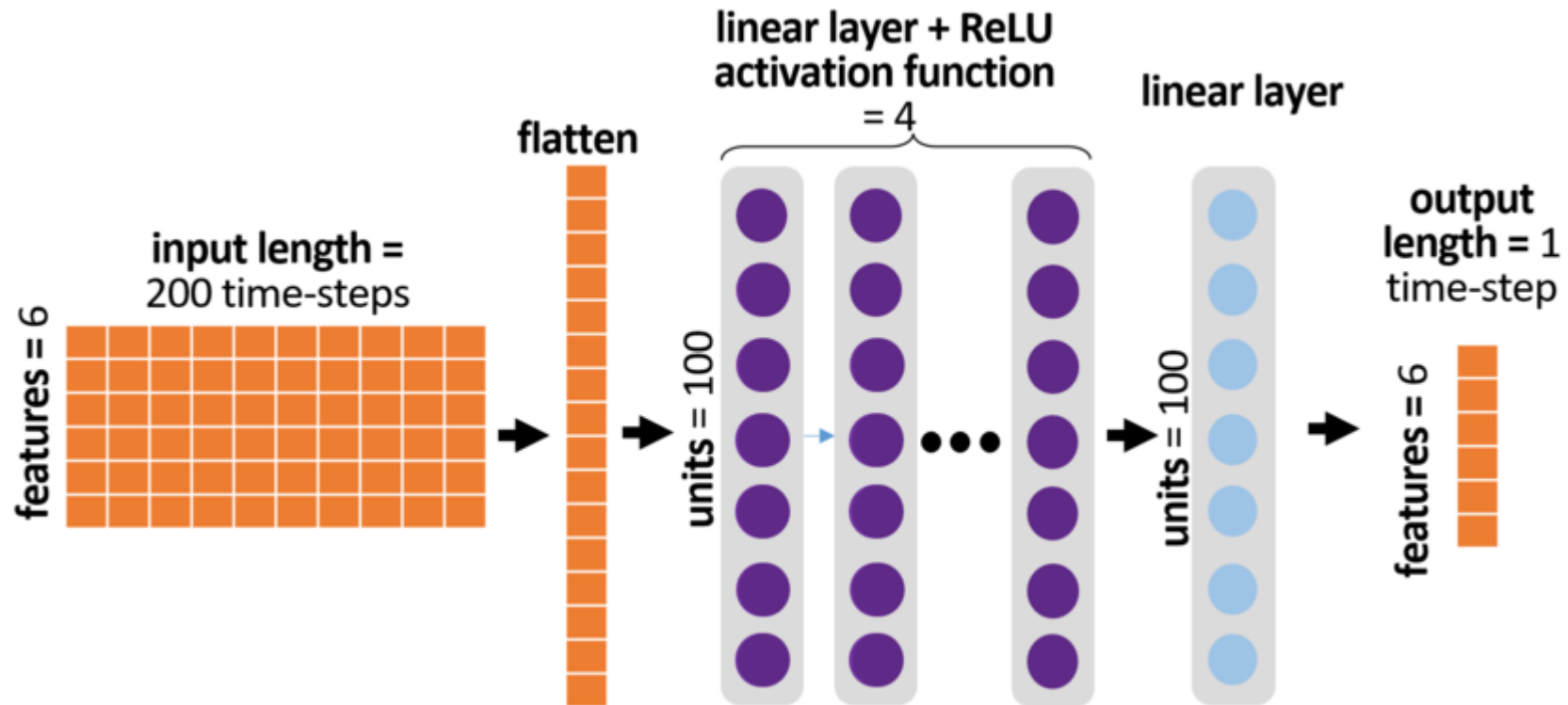
COMPARING CT, CNN AND FCNN



CONVOLUTIONAL NEURAL NETWORKS



FULLY CONNECTED NEURAL NETWORK



Key Features	Convolutional Table	Traditional CNNs	FCNNs
Architecture	Uses pre-defined or learned lookup tables instead of convolution filters	Layers of convolutions followed by pooling	Fully connected layers only
Feature extraction	Good, learned through crafted lookup patterns	Excellent at capturing spatial hierarchies such as edges of an image.	Weak spatial awareness
Data Augmentation	Moderate benefit; depends on table generalization	Highly benefits from augmentation	Minor improvement
Training Complexity	Low (can be trained with simple rules or layer-wise learning)	Medium to High	Lower than CNN, but overfits easily
Efficiency on CPU/ GPU	Extremely fast on CPU, ideal for low-power devices	Works well on GPU	Less GPU efficient
Use Case Examples	Hardware-optimized apps	Image classification, object detection, medical imaging	Simple classification/regression

COMPARISON OF PROGRESS

CIFAR-10 DATASET

PLATFORM	CT	CNN	FCNN
PYTHON(Jupyter Notebook/Marimo)	Yes	Yes	Yes
MATLAB	Yes	Yes	Yes

SVHN and MNIST DATASET

PLATFORM	CT	CNN	FCNN
PYTHON(Jupyter Notebook)	Yes	Yes	Yes

1.) CIFAR-10 DATASET

RESULTS OF CONVOLUTIONAL TABLES

(IN PYTHON, MATLAB, MARIMO,
WEKA)

PYTHON (Jupyter Notebook)

Pred: airplane
True: airplane



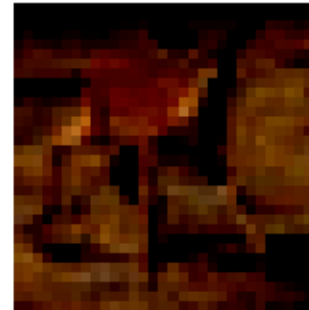
Pred: truck
True: truck



Pred: cat
True: frog



Pred: deer
True: deer



Pred: deer
True: deer



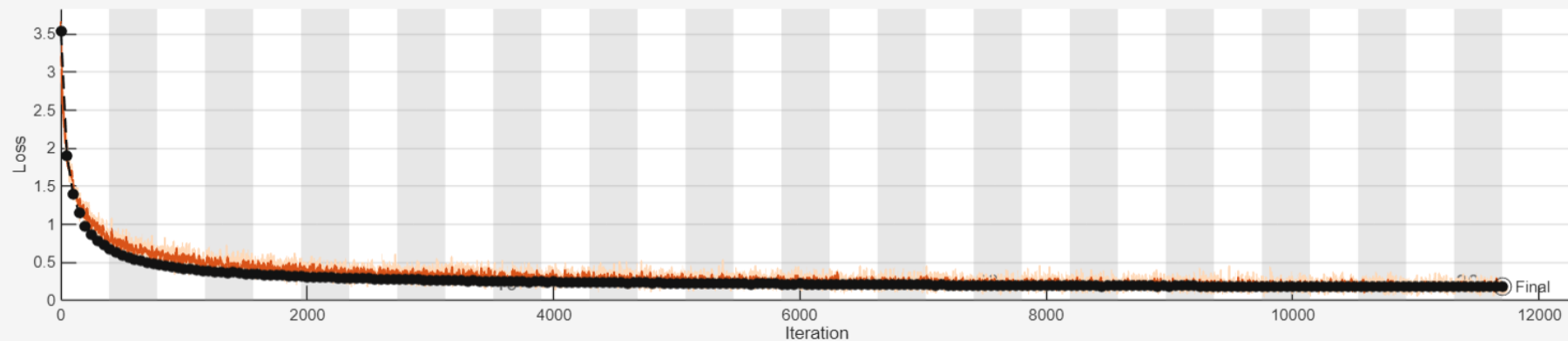
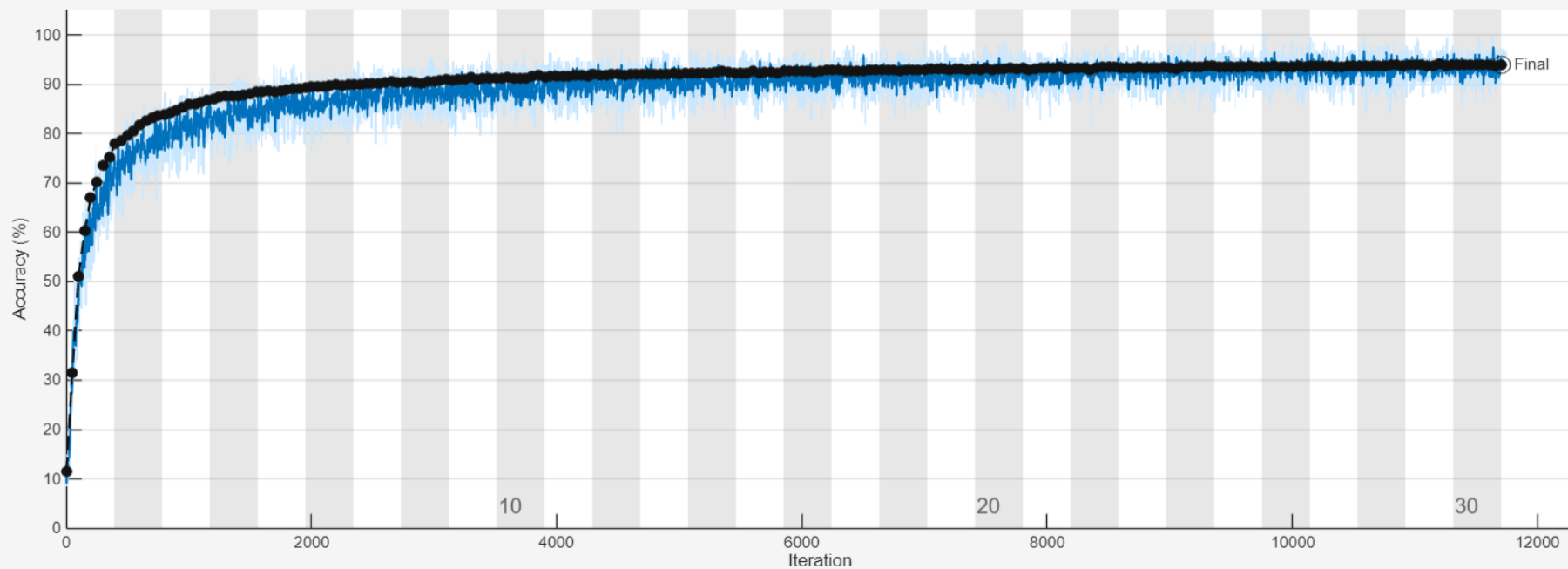
```
Standard Test Loss: 0.4811, Test Accuracy: 83.54%  
Evaluating with test-time augmentation...  
TTA Test Loss: 0.4672, Test Accuracy: 83.65%
```

PYTHON-MARIMO



MATLAB

Training Progress (02-Apr-2025 14:25:03)



Results

Validation accuracy: 93.97%
Training finished: Max epochs completed

Training Time

Start time: 02-Apr-2025 14:25:03
Elapsed time: 5432 min 45 sec

Training Cycle

Epoch: 30 of 30
Iteration: 11700 of 11700
Iterations per epoch: 390
Maximum iterations: 11700

Validation

Frequency: 50 iterations

Other Information

Hardware resource: Single CPU
Learning rate schedule: Constant
Learning rate: 0.0001

Export as Image

[Learn more](#)

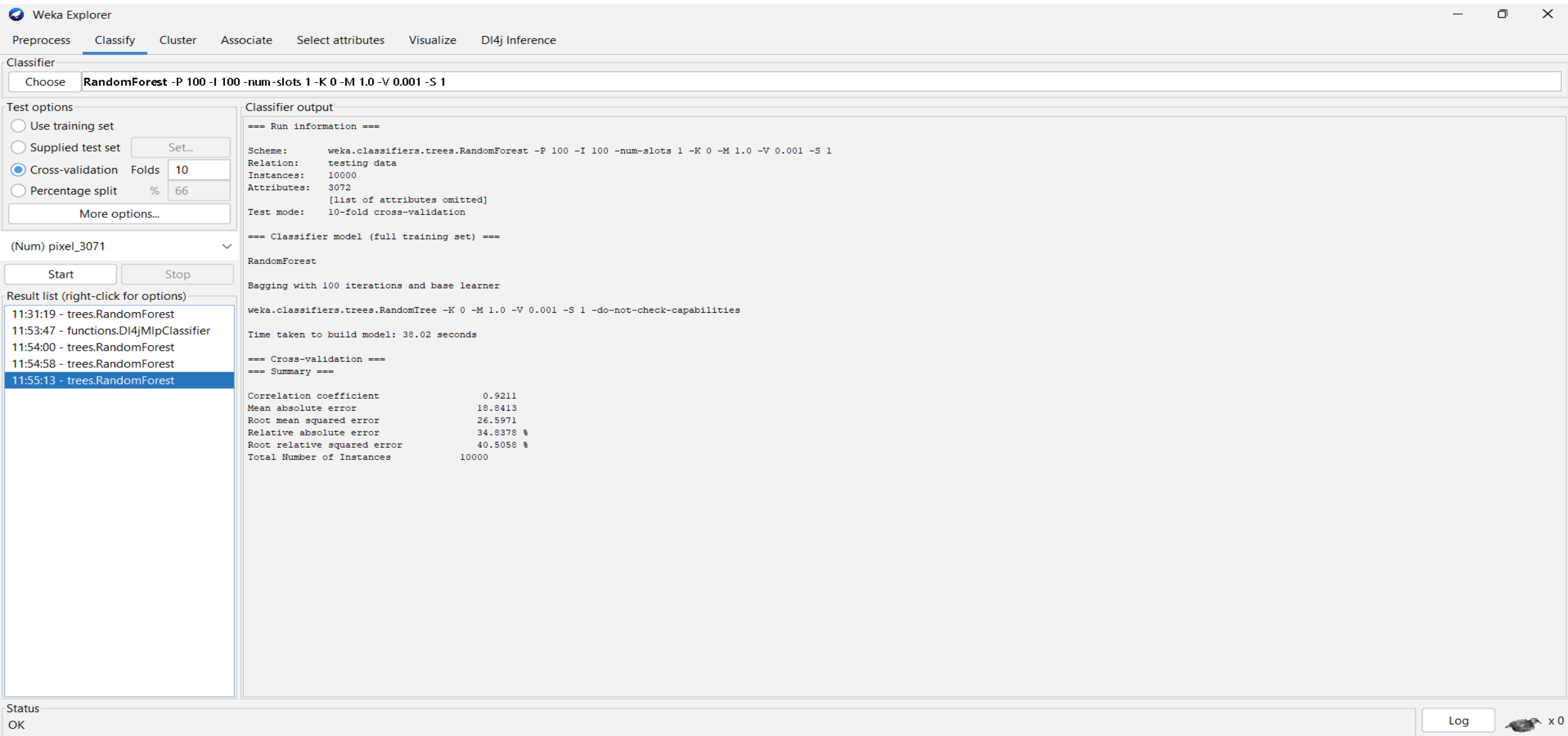
Accuracy

— Training (smoothed)
— Training
— Validation

Loss

— Training (smoothed)
— Training
— Validation

WEKA (CIFAR 10)



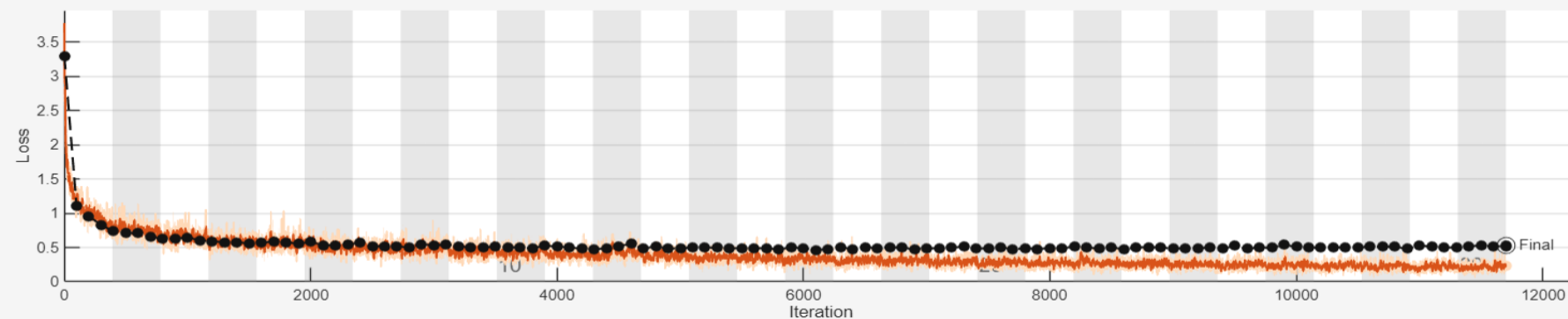
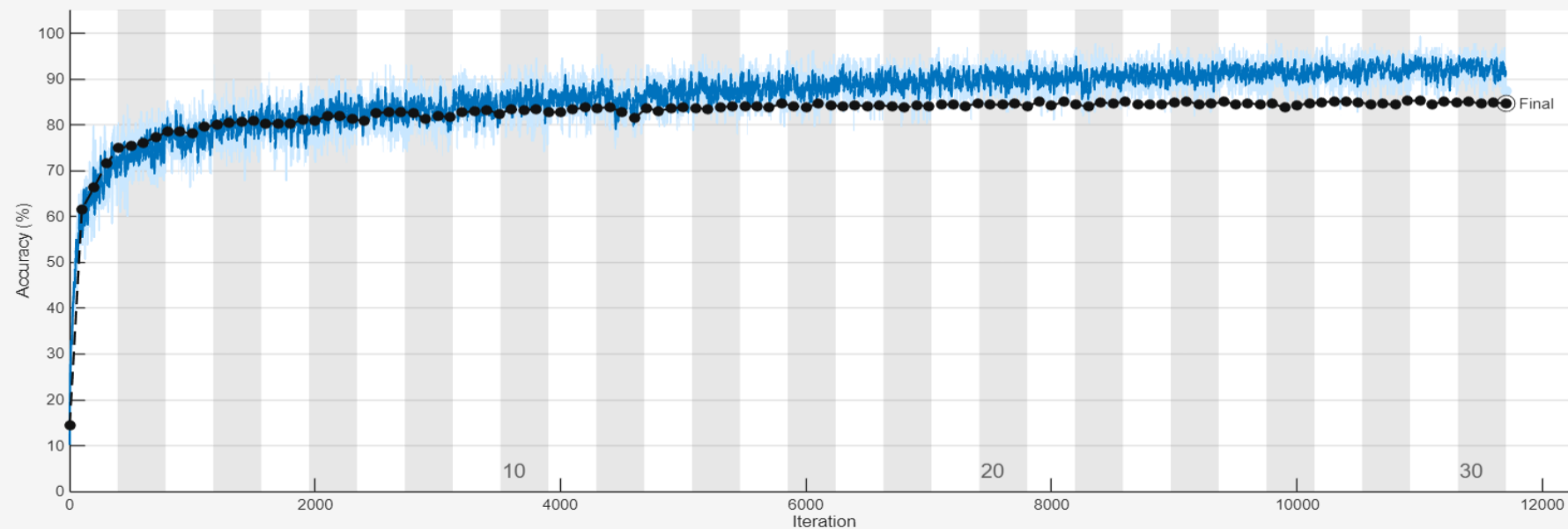
RESULTS OF CONVOLUTIONAL NEURAL NETWORK

(IN MATLAB, PYTHON AND MARIMO)

MATLAB

Training Progress (13-Apr-2025 18:25:01)

Training Progress (13-Apr-2025 18:25:01)



Results

Validation accuracy: 84.63%
Training finished: Max epochs completed

Training Time

Start time: 13-Apr-2025 18:25:01
Elapsed time: 450 min 5 sec

Training Cycle

Epoch: 30 of 30
Iteration: 11700 of 11700
Iterations per epoch: 390
Maximum iterations: 11700

Validation

Frequency: 100 iterations

Other Information

Hardware resource: Single CPU
Learning rate schedule: Piecewise
Learning rate: 0.01

Export as Image

[Learn more](#)

Accuracy

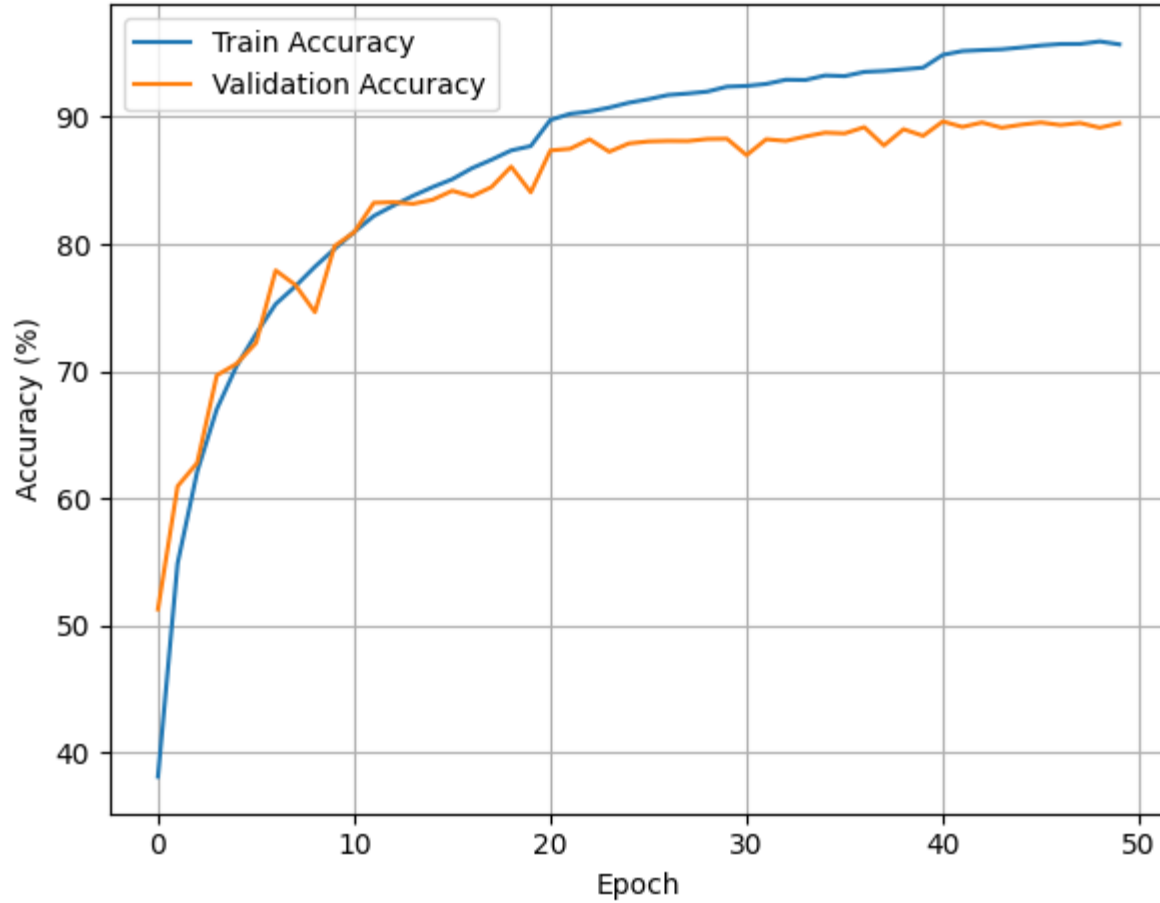
— Training (smoothed)
— Training
— Validation

Loss

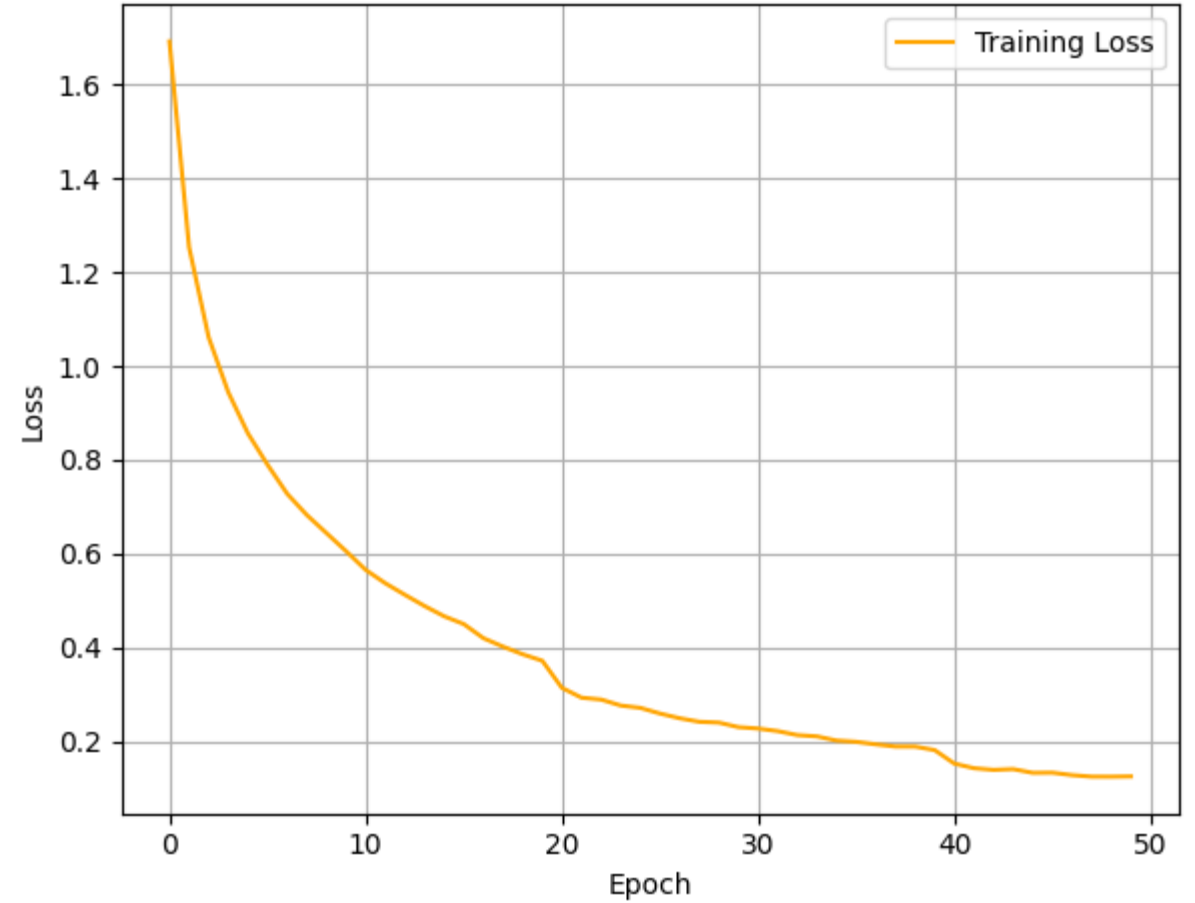
— Training (smoothed)
— Training
— Validation

PYTHON (Jupyter Notebook)

Accuracy over Epochs

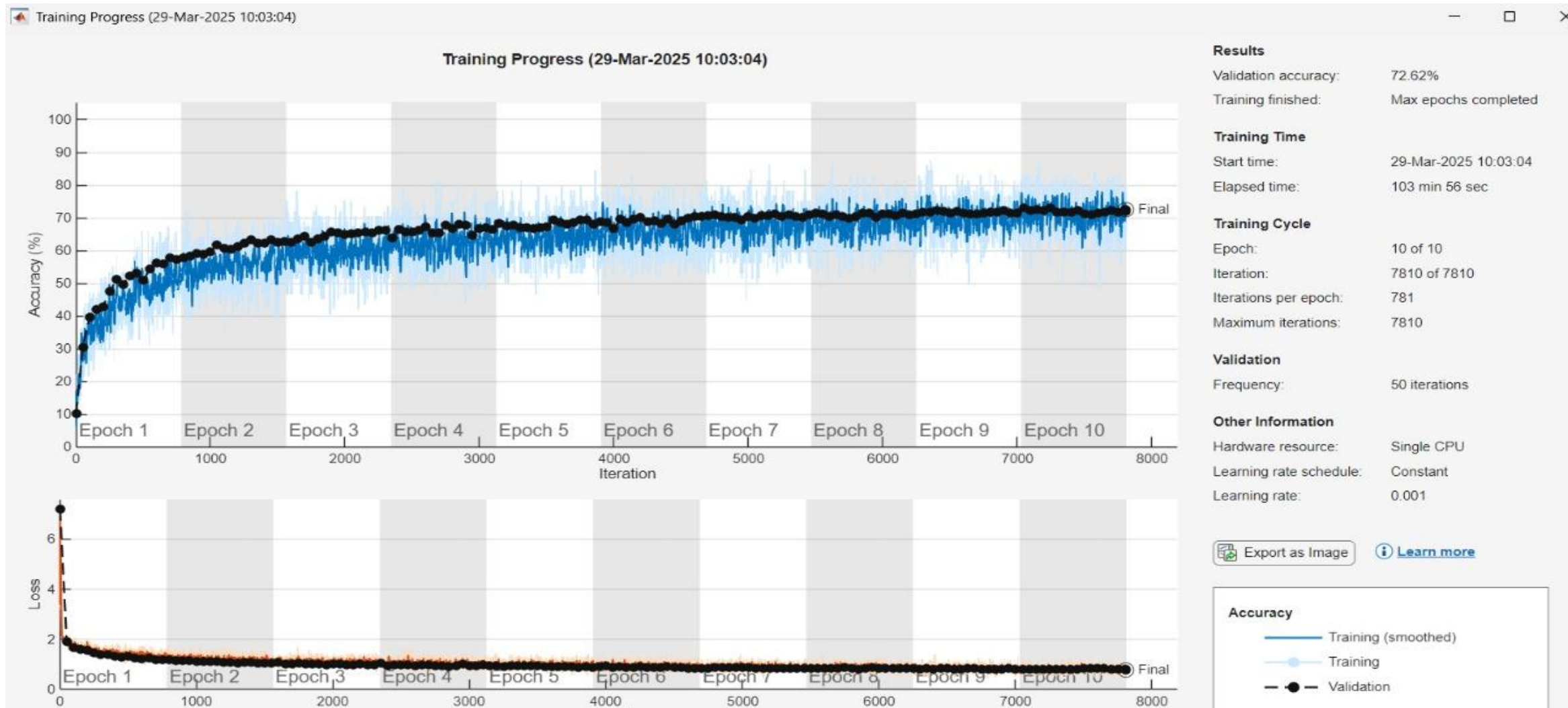


Training Loss over Epochs

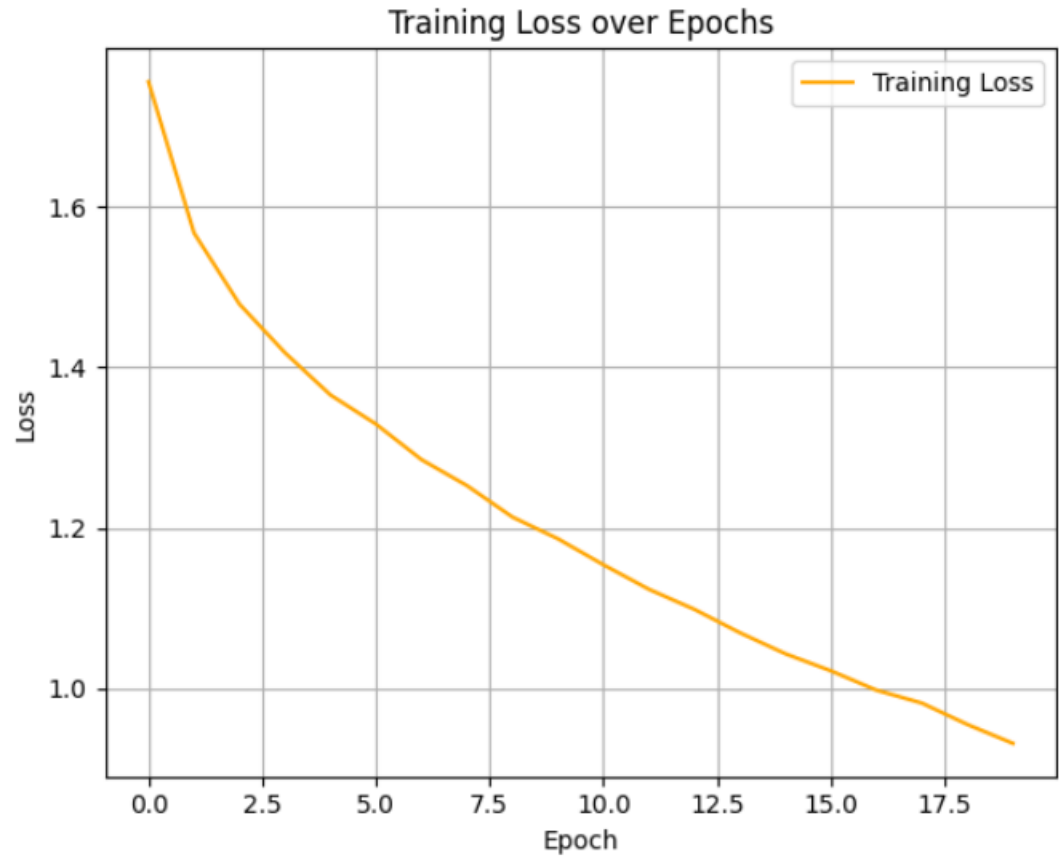
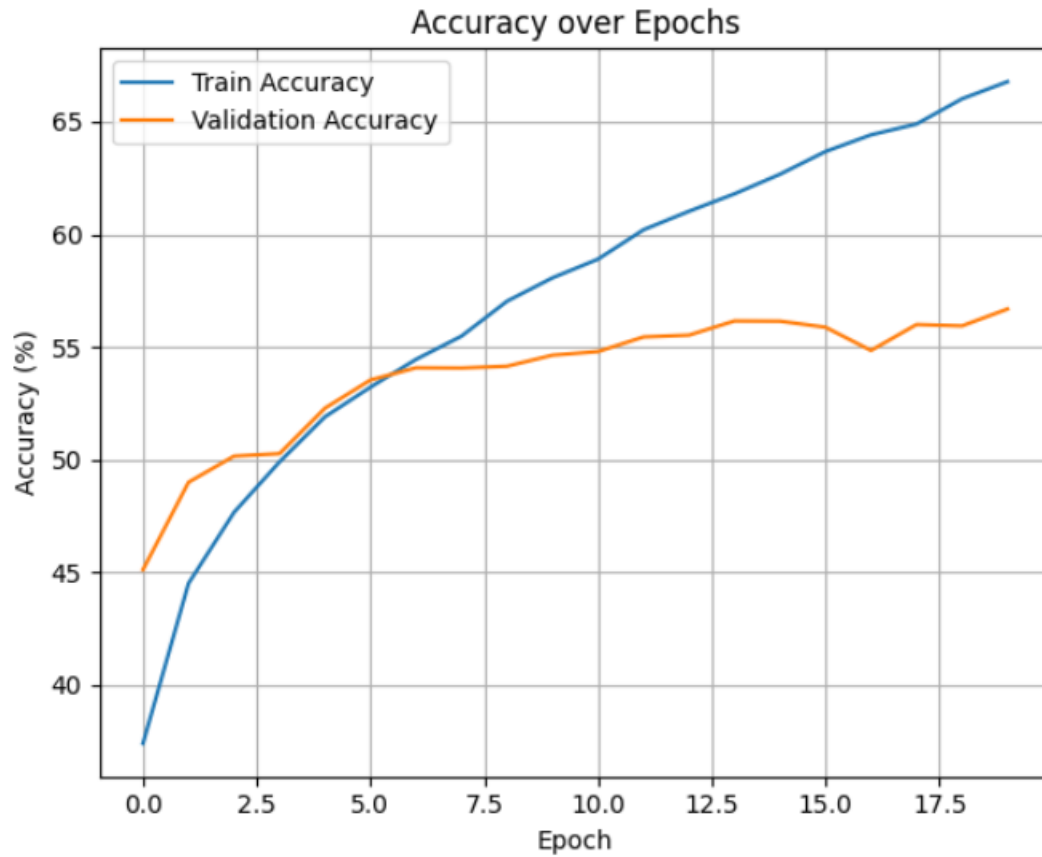


RESULTS OF FULLY CONNECTED NEURAL NETWORKS IN MATLAB

MATLAB



PYTHON (Jupyter Notebook)



2.) MNIST DATASET

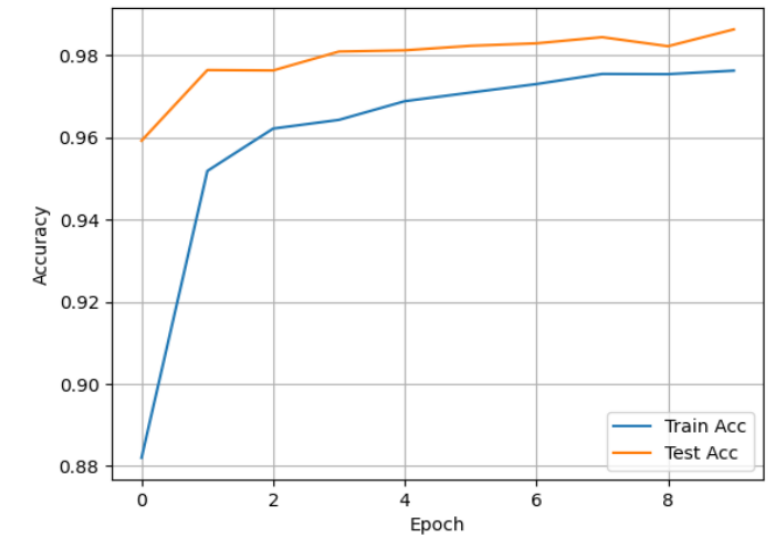
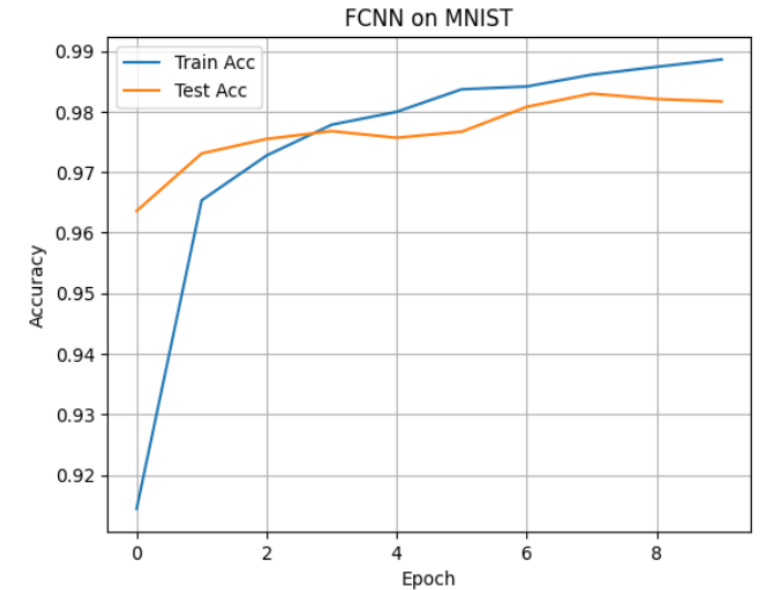
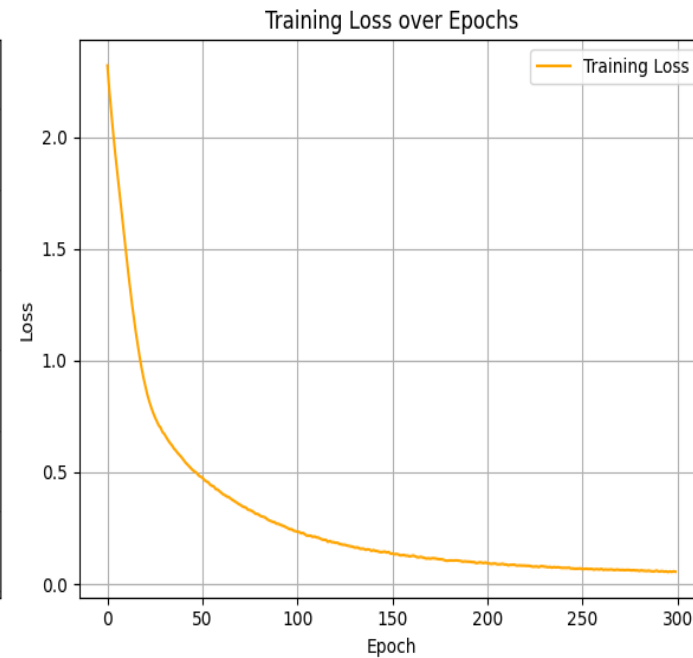
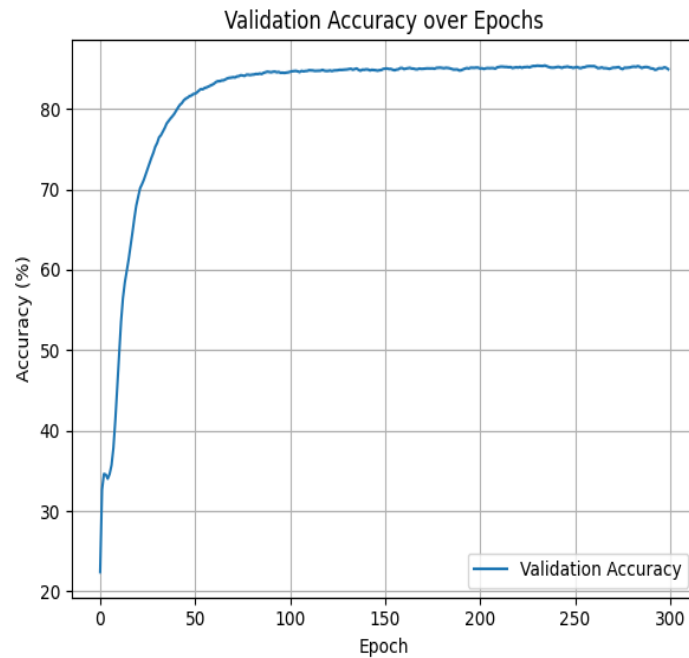
RESULTS OF CTs, CNN and FCNN

(IN PYTHON) and WEKA results

PYTHON (Jupyter Notebook)

➤ Training Progress of CNN and FCNN

➤ Training Progress of CT



WEKA (MNIST DATASET)

=== Summary ===

Correlation coefficient	0.9067
Mean absolute error	21.3835
Root mean squared error	29.3478
Relative absolute error	39.4244 %
Root relative squared error	44.59 %
Total Number of Instances	8000

3.) SVHN DATASET

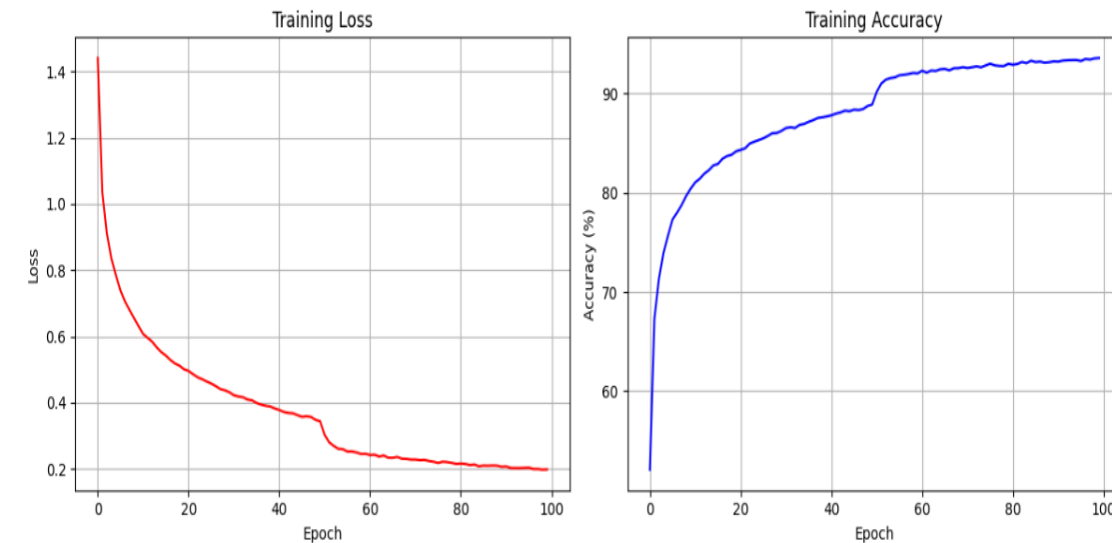
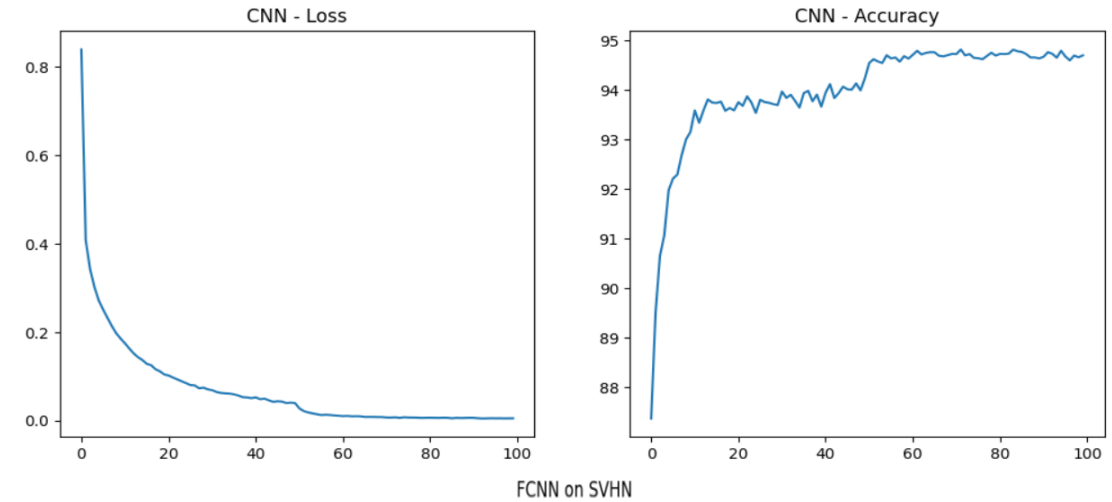
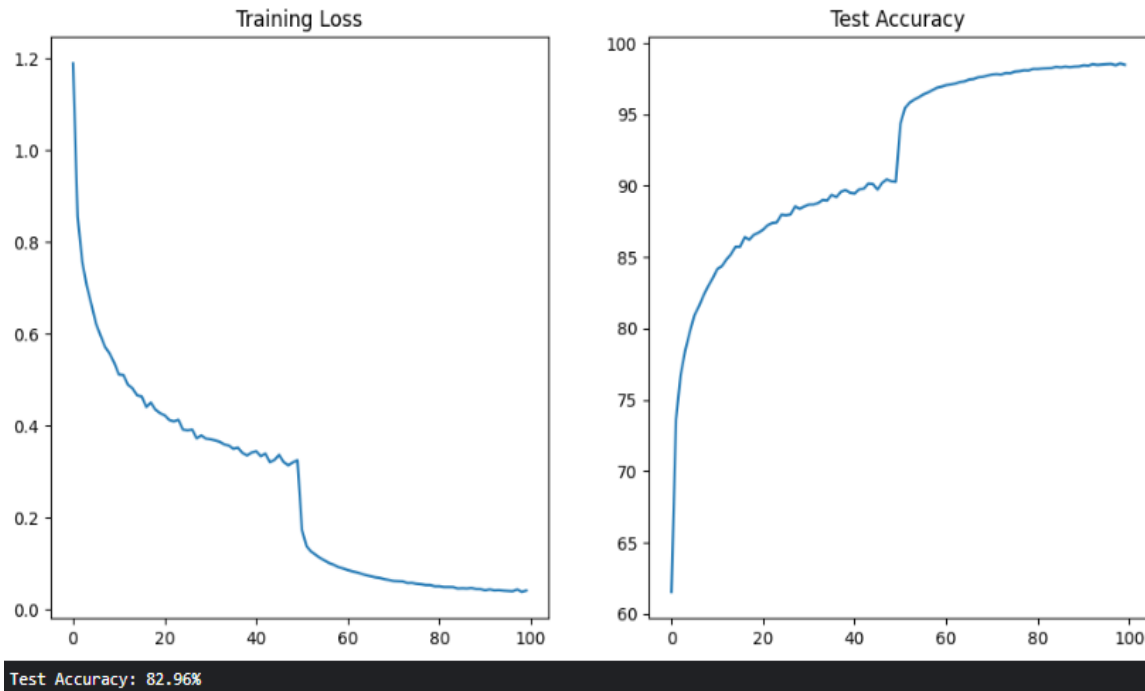
RESULTS OF CTs, CNN and FCNN

(IN PYTHON) and WEKA results

PYTHON (Jupyter Notebook)

➤ Training Progress of CNN and FCNN

➤ Training Progress of CT



Test Accuracy: 86.74%

RESULTS – UI

Simple UI to display results using HTML and CSS

IMPLEMENTATION OF UI



Flask

- Frontend using HTML and CSS to display model results and dataset.
- Connected the UI to a Flask backend to fetch and serve data dynamically from a local SQLite database / implementation of the UI website running through WAMP server.
- The website contains home, introduction, dataset, methodology, results, conclusion, and a reference page respectively as displayed in the upcoming slides.

Deep Learning Model Comparison

CNNs vs FCNNs vs Convolution Tables

[Home](#)[Introduction](#)[Dataset](#)[Methodology](#)[Results](#)[Conclusion](#)

Welcome to Our Deep Learning Project

This research compares the performance of traditional Convolutional Neural Networks (CNNs), Fully Connected Neural Networks (FCNNs), and an innovative approach using Convolution Tables (CTs) on the CIFAR-10 and CIFAR-100 datasets.

Sample CIFAR-10 Images

The CIFAR-10 dataset consists of 60,000 32x32 color images across 10 classes. Below are samples from each class used in our experiments.



Deep Learning Model Comparison

CNNs vs FCNNs vs Convolution Tables

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Project Overview

This deep learning project explores the comparative performance of traditional Convolutional Neural Networks (CNNs), Fully Connected Neural Networks (FCNNs), and an alternative technique called Convolution Tables (CTs). We applied all models to the CIFAR-10 and CIFAR-100 datasets to evaluate their effectiveness in image classification tasks.

Research Objectives

The goal of this research is to highlight the strengths and limitations of each method and investigate whether convolution tables offer a viable, efficient alternative to CNNs in certain use cases. Our analysis focuses on accuracy, computational efficiency, and model complexity as key performance indicators.

Significance

As deep learning continues to evolve, exploring alternative architectures beyond traditional CNNs becomes increasingly important. This study contributes to the ongoing discussion about optimal neural network structures for image classification tasks, particularly in resource-constrained environments where computational efficiency is crucial.

Deep Learning Model Comparison

CNNs vs FCNNs vs Convolution Tables

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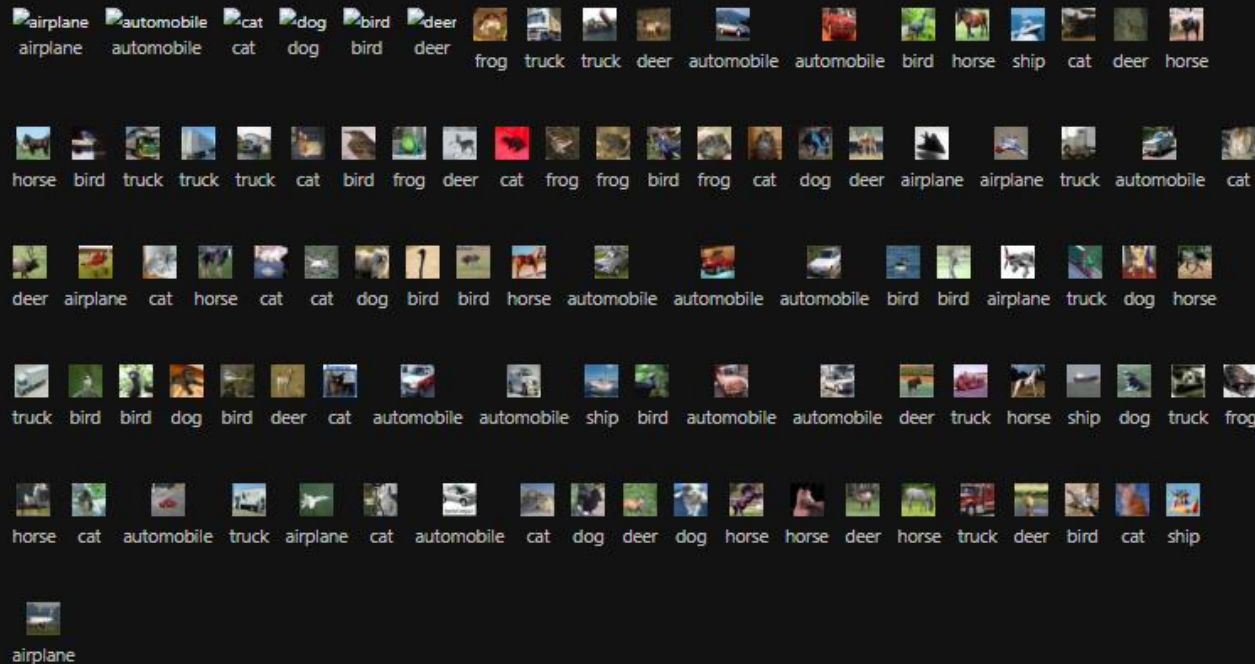
CIFAR-10 and CIFAR-100

The CIFAR-10 and CIFAR-100 datasets consist of colored images of size 32x32 pixels. These datasets are widely used for benchmarking image classification algorithms due to their standardized format and balanced class distribution.

CIFAR-10 Dataset

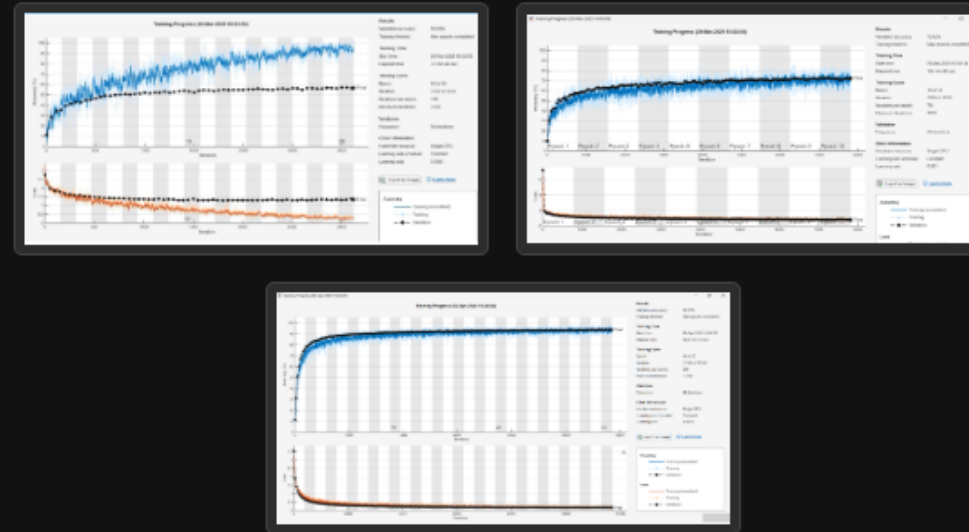
CIFAR-10 contains 60,000 color images across 10 classes, with 6,000 images per class. The training set contains 50,000 images, while the testing set contains 10,000 images. The classes include airplanes, automobiles, birds, cats, deer, dogs, frogs, horses, ships, and trucks.

Live CIFAR-10 Sample Images



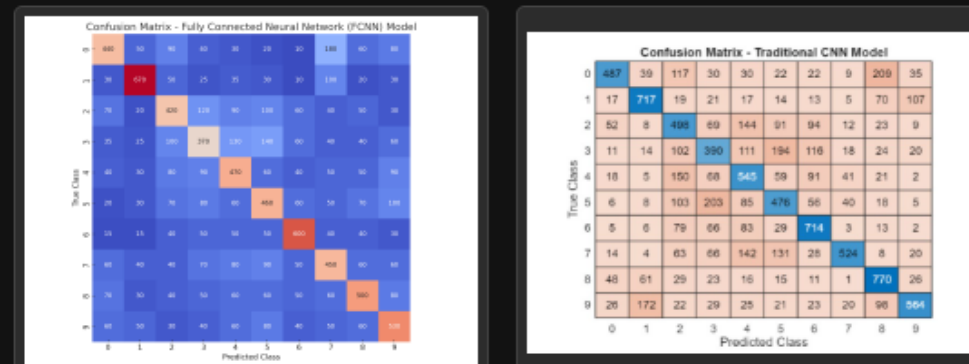
Performance Comparison

Our experiments revealed interesting performance patterns across the three architectures. The traditional CNN achieved 89% accuracy on CIFAR-10, maintaining its reputation for strong performance in image classification tasks. The FCNN reached 82%, while our Convolution Table implementation achieved a promising 87% accuracy.



Classification Accuracy

The confusion matrices below provide deeper insight into each model's classification performance, revealing where each architecture excels and struggles with particular classes.



Key Findings

Our comprehensive analysis revealed that while traditional CNNs still maintain their position as the top performer for image classification tasks, the Convolution Tables approach shows remarkable promise with only a 2% accuracy gap despite its architectural simplicity. FCNNs, while less accurate overall, demonstrated competitive performance on certain classes that have distinctive global features.

Model Analysis

The confusion matrices indicate that CNNs excel at capturing local features that distinguish similar objects, while CTs offer a good balance between global and local feature extraction. FCNNs struggled with classes that rely heavily on spatial relationships but performed well when color distribution and overall shape were sufficient discriminators.

Future Directions

The promising performance of Convolution Tables suggests several interesting avenues for future research. Hybrid architectures that combine the strengths of CNNs and CTs could potentially achieve better performance with lower computational costs. Additionally, exploring the application of these models to other computer vision tasks beyond classification would help establish their broader utility in the deep learning ecosystem.

Deep Learning Model Comparison

CNNs vs FCNNs vs Convolution Tables

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References

Convolutional Tables (CT) Paper:

S. Dekel, Y. Keller and A. Bar-Hillel, "Deep Convolutional Tables: Deep Learning Without Convolutions," in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 10, pp. 13658–13670, Oct. 2024.

[DOI: 10.1109/TNNLS.2023.3270402](https://doi.org/10.1109/TNNLS.2023.3270402)

CIFAR Dataset:

<https://www.cs.toronto.edu/~kriz/cifar.html>

CONCLUSION

- This project explored deep learning without traditional convolutions by implementing **Convolutional Tables (CTs)** with lookup filters and comparing them against standard CNNs and **Fully Connected Neural Networks (FCNNs)** on the CIFAR-10 dataset.
- While CNNs usually achieve the highest accuracy due to their powerful feature learning capabilities, CTs also demonstrated competitive performance with significantly lower computational cost.
- The results highlight CTs as a promising alternative for lightweight and efficient deep learning, especially in resource-constrained environments.

REFERENCES

1. Convolutional Tables (CT) Paper: S. Dekel, Y. Keller and A. Bar-Hillel, "Deep Convolutional Tables: Deep Learning Without Convolutions," in IEEE Transactions on Neural Networks and Learning Systems, vol. 35, no. 10, pp. 13658-13670, Oct. 2024, doi: 10.1109/TNNLS.2023.3270402.
2. <https://www.cs.toronto.edu/~kriz/cifar.html>
3. <https://www.kaggle.com/datasets/stanfordu/street-view-house-numbers>
4. <https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>

THANK YOU